

2020

Full Marks : 70

Time : 3 hours

The figures in the right-hand margin indicate marks

Answer from **all** the Parts as directed

Part—A

(Objective Type Questions)

1. Choose the correct answer in each of the following : 5×1

(a) In digital circuits, shift register circuits is a cascade of

(i) flip-flop (ii) memory

(iii) SP (iv) None of these

(b) Demultiplexer is also known as

(i) Demux

(ii) A/D converter

(iii) Multiple

(iv) None of these

(2)

(c) Half-adder first adds

- (i) two bits (ii) three bits
 - (iii) four bits (iv) None of these
- (d) Which is the microprocessor (μP) comprises?

- (i) Register Section
- (ii) ALU
- (iii) Control Unit
- (iv) All of these

(e) Associative Memory is also known as

- (i) Associative Storage
- (ii) Neural Network
- (iii) Cache Memory
- (iv) None of these

2. Fill in the blanks :

5 × 1

(a) INTR : it implies the _____ signal.

(b) In Register org. PC stands for _____.

(c) DMA stands for _____.

(d) Encoder is reverse form of _____.

(e) The size of each sequents in 8086 is _____.

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(Continued)

(3)

Part—B

(Short Answer Type Questions)

3. Answer *any four* questions of the following : 4×5

- (a) What is instruction cycle? Explain the types of instruction cycle.
- (b) Explain the microprogrammed control with diagram.
- (c) Differentiate between half-adder and full-adder.
- (d) What is mapping? Explain the types of mapping in CSA.
- (e) Differentiate between real mode and protected mode.
- (f) Write short notes on *any two* of the following :
 - (i) Addressing modes
 - (ii) Associative memory
 - (iii) Ports
 - (iv) Pipelining
 - (v) Parallel processing.

(4)

Part—C

(Long Answer Type Questions)

4. Answer any four questions of the following : 4×10

- (a) What is Bus System? Explain the Bus Architecture and Organisation with diagram.
- (b) What is Register? Describe the Register Organisation with diagram.
- (c) Differentiate between ISA and EISA Bus Architecture.
- (d) Explain the Architecture and Organisation of 8086 microprocessors (μP).
- (e) Differentiate between DX and SX of different microprocessor (μP).
- (f) Write short notes on any three of the following :
 - (i) DMA controller
 - (ii) Parallel processing
 - (iii) Instruction set
 - (iv) Cache memory
 - (v) Core if
 - (vi) Clock pulse generator.

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BCA (Sem II) - CC 04

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